Similarity and normalisation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

For query [3, 4] and candidate [3, 0], calculate squared Euclidean distance. Does a smaller distance prove equivalent meaning?

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Calculate vector length and normalise a nonzero vector. Try the recall before opening notes.

Your learning target

I can calculate **vector length** and normalise a nonzero vector.

I can compare squared distance with **cosine similarity** and explain a changed ranking.

I can handle zero vectors explicitly and justify what information a comparison keeps.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Embeddings and neighbourhoods. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Augmentation and leakage. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Convolution as a local operation. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Calculate **vector length** and normalise a nonzero vector

Compare squared distance with **cosine similarity** and explain a changed ranking

Handle zero vectors explicitly and justify what information a comparison keeps

Retrieve what you already know

> For query [3, 4] and candidate [3, 0], calculate squared Euclidean distance. Does a smaller distance prove equivalent meaning?

My starting explanation

This week's end goal

By the end, I can explain similarity and normalisation using a traced example and a stated limitation.

Notice the evidence

> Investigate the supplied Similarity and normalisation case. Separate observations, given facts and assumptions before choosing an explanation.

Last week, a neighbourhood depended on a chosen distance rule. This week, compare the same fictional vectors in two ways. Squared distance rewards nearby coordinates.

Cosine similarity compares direction and ignores positive changes in length. Neither rule is automatically right for every task.

Vector	Coordinates	Length
Query Q	[3, 4]	5
Candidate A	[6, 8]	10
Candidate B	[3, 0]	3
Candidate C	[-3, -4]	5

For Q, length = square root of (3 squared + 4 squared) = square root of 25 = 5. Divide both components by 5 to get its **unit vector** [0.6, 0.8]. Its length is one because 0.6 squared + 0.8 squared = 1.

A becomes the same **unit vector**. C becomes [-0.6, -0.8], pointing in the opposite direction.

These supplied coordinates are for arithmetic practice. They are not learned language embeddings. Normalising each row keeps its direction but removes its original length unless that length is saved separately.

> **List the given facts and one fact that is not supplied.**

See how it works

> Follow the mechanism for Similarity and normalisation. Connect each labelled input, deciding step and result.

Multiply matching components and add to obtain the dot product. **Cosine similarity** equals dot(Q, candidate) divided by (length(Q) times length(candidate)). Higher cosine means a closer direction.

For A, the dot product is 3 times 6 + 4 times 8 = 50, so cosine = 50 / (5 times 10) = 1. For B it is 9 / 15 = 0.6; for C it is -25 / 25 = -1.

```
from math import sqrt

def unit(v):
    if not v:
        raise ValueError("Need a nonempty vector")
    length = sqrt(sum(x*x for x in v))
    if length == 0:
        raise ValueError("Zero vector has no direction")
    return [x / length for x in v]

def cosine(a, b):
    if len(a) != len(b):
        raise ValueError("Need matching dimensions")
    ua, ub = unit(a), unit(b)
    return sum(x*y for x, y in zip(ua, ub))

q = [3, 4]
for name, v in [("A", [6, 8]), ("B", [3, 0]),
                ("C", [-3, -4])]:
    print(name, round(cosine(q, v), 6))
```

Trace before running: A gives 1, B gives 0.6 and C gives -1. This classroom code assumes finite, moderate numeric **input**s. It rejects empty, unequal-length and zero vectors.

Keep full precision for decisions; rounding here is only for display. Cosine lies between -1 and 1 in exact arithmetic and is not a probability.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Similarity and normalisation. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

For query Q, write the raw squared distances and cosines to A, B and C from the lesson. Name the winner under each rule. Calculate squared distances after normalising both sides and check the relationship to cosine.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

Change one thing

Change: Replace P with [0, 0].

Predicted result

Prediction: Cosine and unit normalisation for P are undefined.

Why the result follows

Reason: The zero vector has norm 0, so dividing by its norm is invalid.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Similarity and normalisation to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Vector length	The square root of the sum of squared components, also called the L2 norm
L2 normalisation	Dividing every component of a nonzero vector by its length
Cosine similarity	The dot product of two nonzero vectors divided by the product of their lengths
Unit vector	A vector with length one

Use two terms to explain a step in this week’s example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Similarity and normalisation, predict a step, then compare with its explanation.

Compare the same candidates using two rules

CANDIDATE	RAW SQUARED DISTANCE	COSINE SIMILARITY
A [6, 8]	25	1
B [3, 0]	16	0.6
C [-3, -4]	100	-1

For Q [3, 4], raw squared distance selects B by minimising the score. Cosine selects A by maximising the score. The candidates stayed the same; the comparison rule changed.

After normalising both Q and every candidate, squared distances to A, B and C are 0, 0.8 and 4. For **unit vector**s, squared distance = 2 - 2 times cosine.

Thus the smallest normalised distance and largest cosine produce the same ranking, including ties, in exact arithmetic.

> **Add candidate D [0, 5]. Calculate its raw squared distance and cosine from Q [3, 4]. Compare D with A under each rule.**

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Similarity and normalisation

Compare angular alignment with raw distance on the same vectors.

The facts we will use

Supplied facts

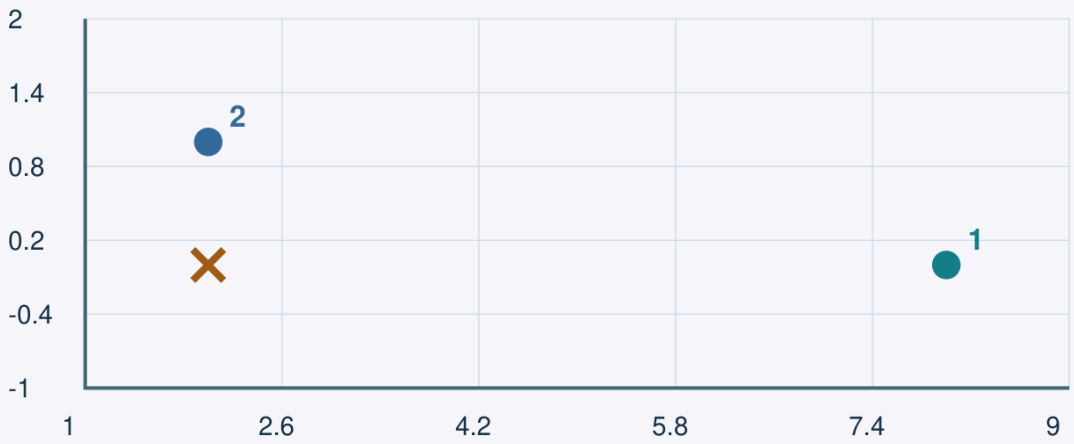
Query [2, 0]; P=[8, 0]; Q=[2, 1]. Nonzero vectors only; compare squared Euclidean distance and cosine.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/27

Raw proximity differs from angular alignment



Horizontal: first vector coordinate; vertical: second vector coordinate

1: P at (8, 0)

2: Q at (2, 1)

Cross: query at (2, 0)

Distance rule: Compare squared Euclidean distance and cosine

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/27

Similarity and normalisation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Raw distances

P distance 36; Q distance 1, so Q is closer in raw coordinates.



2. Angular similarity

Cosine P =1; cosine Q =2/sqrt5≈0.8944, so P aligns more closely.



3. Normalise

Unit query [1, 0], P [1, 0], Q [2/sqrt5, 1/sqrt5]; normalised distance to P is 0.

CHANGE ONE THING

Replace P with [0, 0].

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Raw distances	P distance 36; Q distance 1, so Q is closer in raw coordinates.
2. Angular similarity	Cosine P =1; cosine Q =2/sqrt5≈0.8944, so P aligns more closely.
3. Normalise	Unit query [1, 0], P [1, 0], Q [2/sqrt5, 1/sqrt5]; normalised distance to P is 0.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Raw Euclidean distance includes differences in both direction and magnitude. Cosine compares direction using the **vector length**s, so a long vector can align perfectly while being far away in raw coordinates.

Declare which question your similarity measure answers. A zero vector has no direction and cannot be divided by its norm for cosine or unit normalisation.

What this example does not establish

Keep the stated limit

Neither metric is universally right; choose it from the task and handle zero vectors explicitly.

Keep the stated limit

Read the labelled picture: Why do the original two metrics select different winners? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Replace P with [0, 0]. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Similarity and normalisation claim and explain a repair using the supplied evidence.

Check the claim against the facts

Suppose [3, 4] records counts of two event types, while [6, 8] records twice as many of each. Both have the same proportions and direction, but totals are 7 and 14. Normalisation makes their vectors identical.

That may suit a question about proportions, but it removes information needed to compare total activity. For learned embeddings, the usefulness of length must be tested for the task.

Check the claim against the facts

The zero vector [0, 0] has no direction, so its cosine is undefined. Do not silently assign zero similarity: zero cosine already means perpendicular nonzero vectors.

This lesson rejects a zero query and excludes zero candidates with an explicit report. If no valid candidates remain, return no comparison. PAD entries stay excluded by the mask; a real zero **input** needs its own policy.

Positive scaling preserves cosine: Q and 2 times Q have the same direction. Negative scaling reverses direction; Q and -Q have cosine -1.

Candidate labels, self-exclusion and a stated floating-point tie tolerance are still required when retrieving neighbours.

Locate the earliest unsupported step

A candidate set contains Z [0, 0] and B [3, 0]. With nonzero query Q, what happens under this lesson's zero-vector policy? What if Q is zero, or Z is the only candidate?

Explain why replacing an undefined cosine with zero is misleading.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Similarity and normalisation with support, then explain the deciding step in your own words.

For Q [3, 4], A [6, 8] and C [-3, -4], give each length and **unit vector**. Explain one distinction that normalisation removes and one that it preserves.

> **Add candidate D [0, 5]. Calculate its raw squared distance and cosine from Q [3, 4]. Compare D with A under each rule.**

> **Explain your prediction to a partner before comparing results.**

Practise with less help: Compare the rankings

Use workbook W2 for the connected example “Similarity and normalisation”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Similarity and normalisation. Proposed explanation: “Cosine is a probability”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why do the original two metrics select different winners? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Similarity and normalisation. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

For query Q, write the raw squared distances and cosines to A, B and C from the lesson. Name the winner under each rule. Calculate squared distances after normalising both sides and check the relationship to cosine.

```
from math import sqrt

def unit(v):
    if not v:
        raise ValueError("Need a nonempty vector")
    length = sqrt(sum(x*x for x in v))
    if length == 0:
        raise ValueError("Zero vector has no direction")
    return [x / length for x in v]
```

```
def cosine(a, b):
    if len(a) != len(b):
        raise ValueError("Need matching dimensions")
    ua, ub = unit(a), unit(b)
    return sum(x*y for x, y in zip(ua, ub))

q = [3, 4]
for name, v in [("A", [6, 8]), ("B", [3, 0]),
                ("C", [-3, -4])]:
    print(name, round(cosine(q, v), 6))
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Similarity and normalisation. Explain why each step is justified.

Explain → apply → challenge

For Q [3, 4], A [6, 8] and C [-3, -4], give each length and **unit vector**. Explain one distinction that normalisation removes and one that it preserves.

A candidate set contains Z [0, 0] and B [3, 0]. With nonzero query Q, what happens under this lesson's zero-vector policy? What if Q is zero, or Z is the only candidate?

Explain why replacing an undefined cosine with zero is misleading.

For event-count vectors [3, 4] and [6, 8], what useful information can normalisation remove? A team chooses cosine because it improves results on the final evaluation set, then calls that same set independent **evidence**.

Explain how to repair the evaluation plan.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled similarity and normalisation **evidence** sheet to Image-pattern Network Investigation. Show how you can calculate **vector length** and normalise a nonzero vector.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

L2 normalisation divides each individual vector by its own length; it does not estimate a mean or scale from a dataset. Feature standardisation is a different operation and must fit its statistics on **training** data only.

Choose between raw distance and cosine using the task, **training** and validation **evidence**. Do not repeatedly change the rule after seeing final evaluation results.

A teammate says a cosine of 1 proves two records mean the same thing, and normalisation always improves a model. Use A and Q to challenge both claims. What must be checked before adopting the rule?

Record the representation version, comparison rule, normalisation, zero-vector handling, candidate set and tie tolerance. Preserve a baseline and evaluate on the same reserved examples.

Next week, transfer learning asks when a representation learned for one task can help another.

For event-count vectors [3, 4] and [6, 8], what useful information can normalisation remove? A team chooses cosine because it improves results on the final evaluation set, then calls that same set independent **evidence**.

Explain how to repair the evaluation plan.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Similarity and normalisation case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Compare the same candidates using two rules

CANDIDATE	RAW SQUARED DISTANCE	COSINE SIMILARITY
A [6, 8]	25	1
B [3, 0]	16	0.6
C [-3, -4]	100	-1

For Q [3, 4], raw squared distance selects B by minimising the score. Cosine selects A by maximising the score. The candidates stayed the same; the comparison rule changed.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Compare a new case independently

Use query [5, 0] and candidates P [10, 0], R [4, 3] and S [0, 5]. Calculate all raw squared distances and cosines, name each winner, and give squared distances after normalising both query and candidates.

Explain the changed ranking.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Use query [5, 0] and candidates P [10, 0], R [4, 3] and S [0, 5]. Calculate all raw squared distances and cosines, name each winner, and give squared distances after normalising both query and candidates.

Explain the changed ranking.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Similarity and normalisation, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Compare angular alignment with raw distance on the same vectors.

By the end, I can	How I will show it
Calculate vector length and normalise a nonzero vector	Explain the deciding step, show a trace or test, and state one limit.
Compare squared distance with cosine similarity and explain a changed ranking	Explain the deciding step, show a trace or test, and state one limit.
Handle zero vectors explicitly and justify what information a comparison keeps	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Similarity and normalisation without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Calculate **vector length** and normalise a nonzero vector

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Compare angular alignment with raw distance on the same vectors.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Unit query [1, 0], P [1, 0], Q [2/sqrt5, 1/sqrt5]; normalised distance to P is 0.

1. Angular similarity

2. Normalise

3. Raw distances

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Normalise Unit query [1, 0], P [1, 0], Q [2/sqrt5, 1/sqrt5]; normalised distance to P is 0. Raw Euclidean distance includes differences in both direction and magnitude.

Cosine compares direction using the **vector length**s, so a long vector can align perfectly while being far away in raw coordinates. Declare which question your similarity measure answers.

A zero vector has no direction and cannot be divided by its norm for cosine or unit normalisation.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Replace P with [0, 0]. Which conclusion is supported by the supplied facts?

1. Cosine and unit normalisation for P are undefined.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Cosine and unit normalisation for P are undefined. The zero vector has norm 0, so dividing by its norm is invalid.

Repair a missing building block

Review Embeddings and neighbourhoods (ai-5/26). Goal: Look up a vector using a token ID and preserve sequence shape. Short practice: What would justify claiming nearby vectors mean related topics?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Augmentation and leakage (ai-5/24). Goal: Check whether a transformation preserves the task label. Short practice: How many independent source groups are in the final test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Convolution as a local operation (ai-5/21). Goal: Compute a local weighted sum at each valid image position. Short practice: Why is flipping the kernel an implementation change here?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

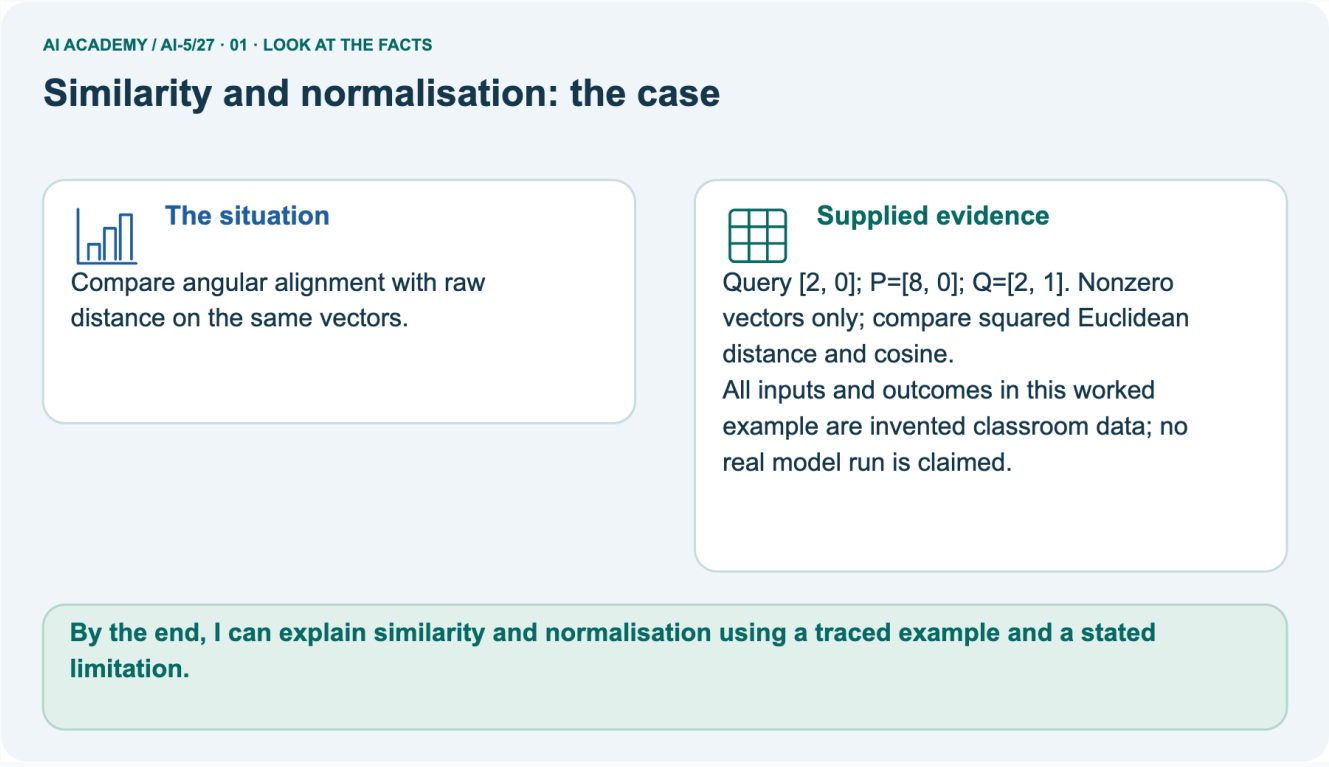
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Similarity and normalisation graphic using its labels, deciding evidence and relevant limitation.

01. Similarity and normalisation: the case



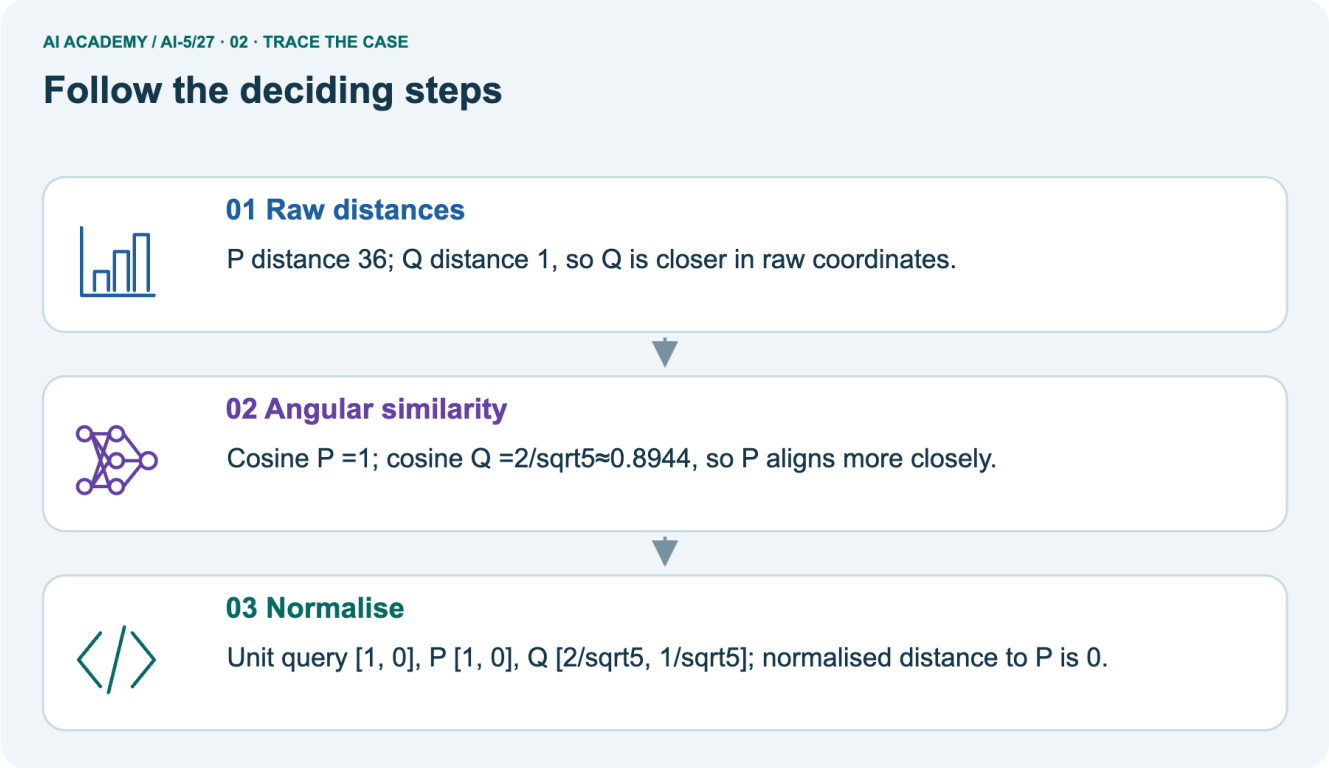
Follow the labelled picture with the matching explanation. Open the vector diagram

Compare angular alignment with raw distance on the same vectors. Supplied facts: Query [2, 0]; P=[8, 0]; Q=[2, 1]. Nonzero vectors only; compare squared Euclidean distance and cosine.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

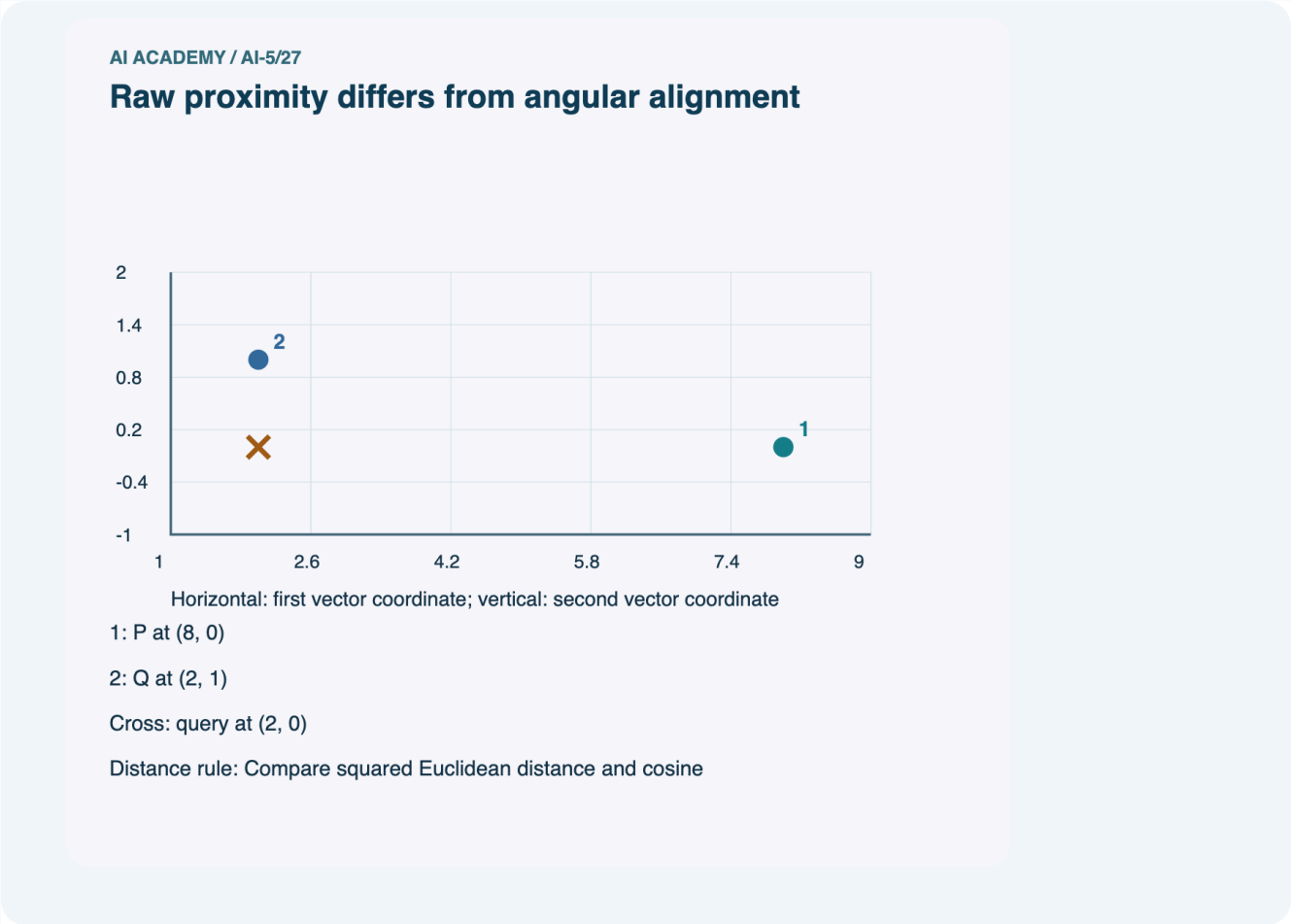


Follow the labelled picture with the matching explanation. Open the vector diagram

Raw distances: P distance 36; Q distance 1, so Q is closer in raw coordinates. → Angular similarity: Cosine P =1; cosine Q =2/sqrt5≈0.8944, so P aligns more closely. → Normalise: Unit query [1, 0], P [1, 0], Q [2/sqrt5, 1/sqrt5]; normalised distance to P is 0.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Raw proximity differs from angular alignment

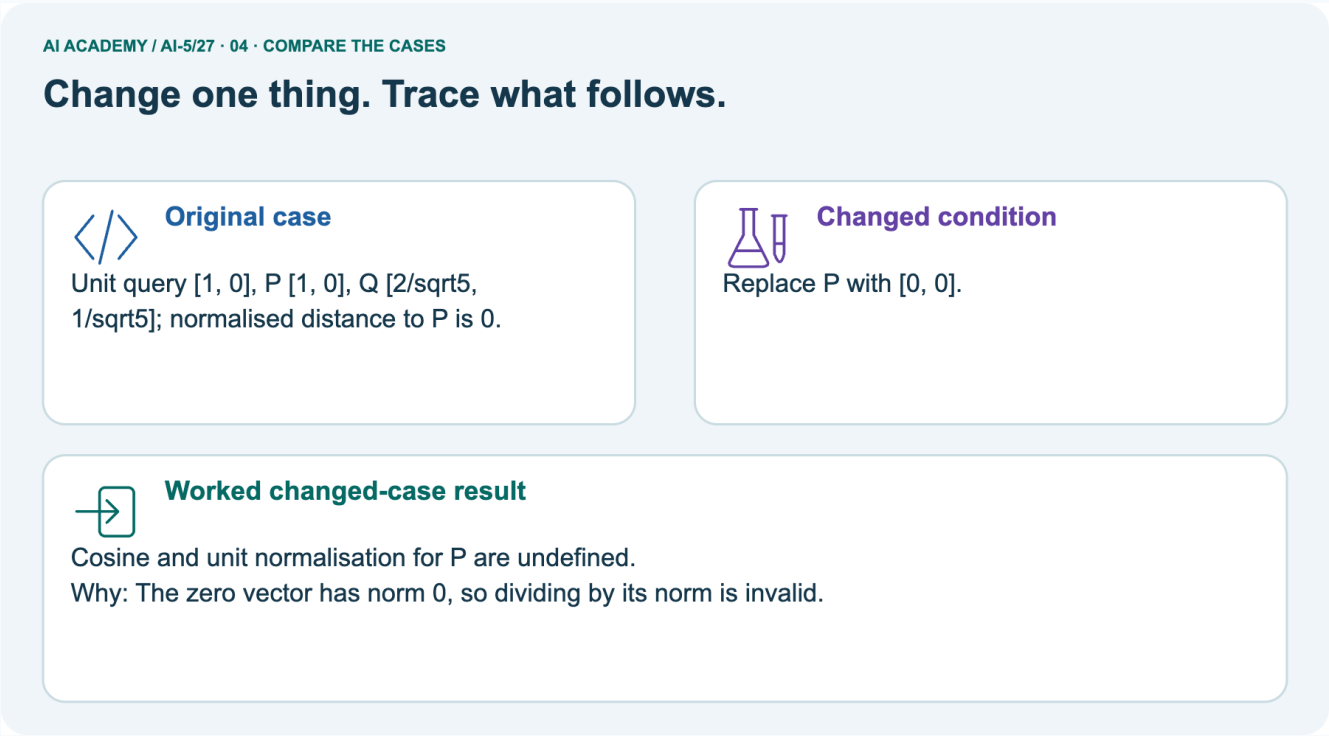


Follow the labelled picture with the matching explanation. Open the vector diagram

Raw proximity differs from angular alignment. P: (8, 0). Q: (2, 1). Query query: (2, 0). Axes: first vector coordinate; second vector coordinate. Distance: Compare squared Euclidean distance and cosine.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



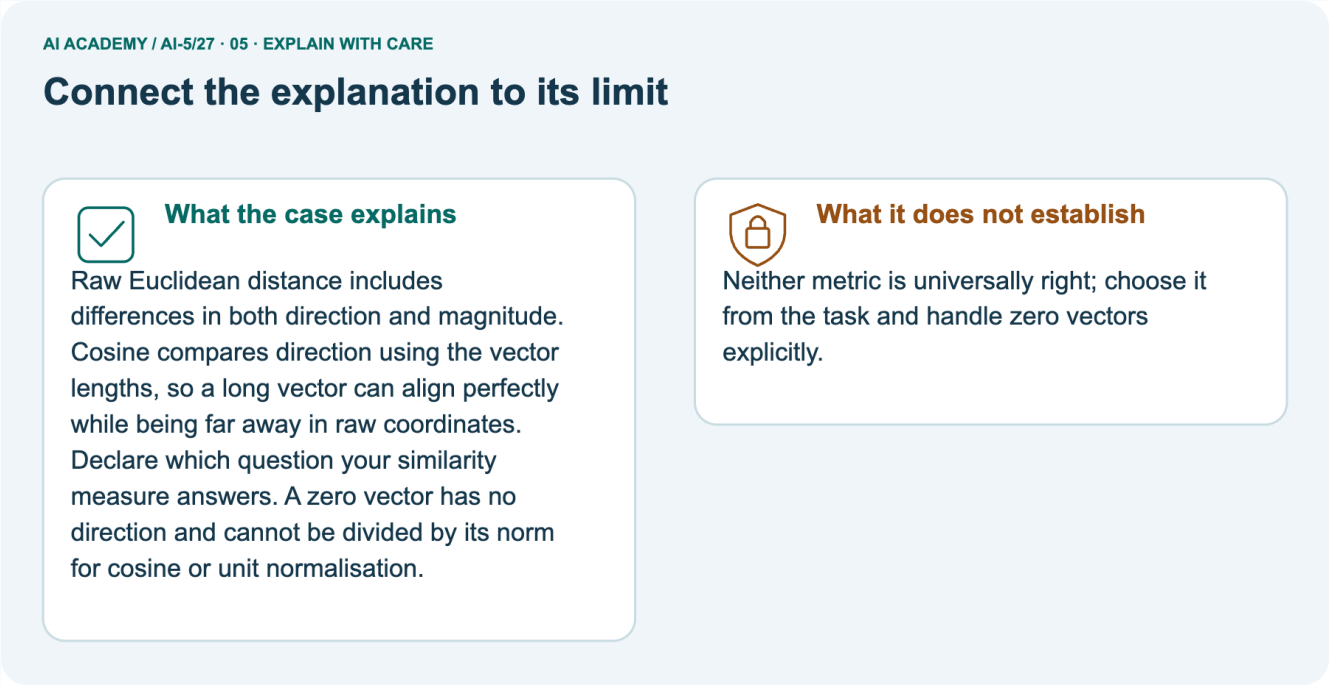
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Unit query [1, 0], P [1, 0], Q [2/sqrt5, 1/sqrt5]; normalised distance to P is 0. Changed condition: Replace P with [0, 0]. Expected result: Cosine and unit normalisation for P are undefined.

Reason: The zero vector has norm 0, so dividing by its norm is invalid.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

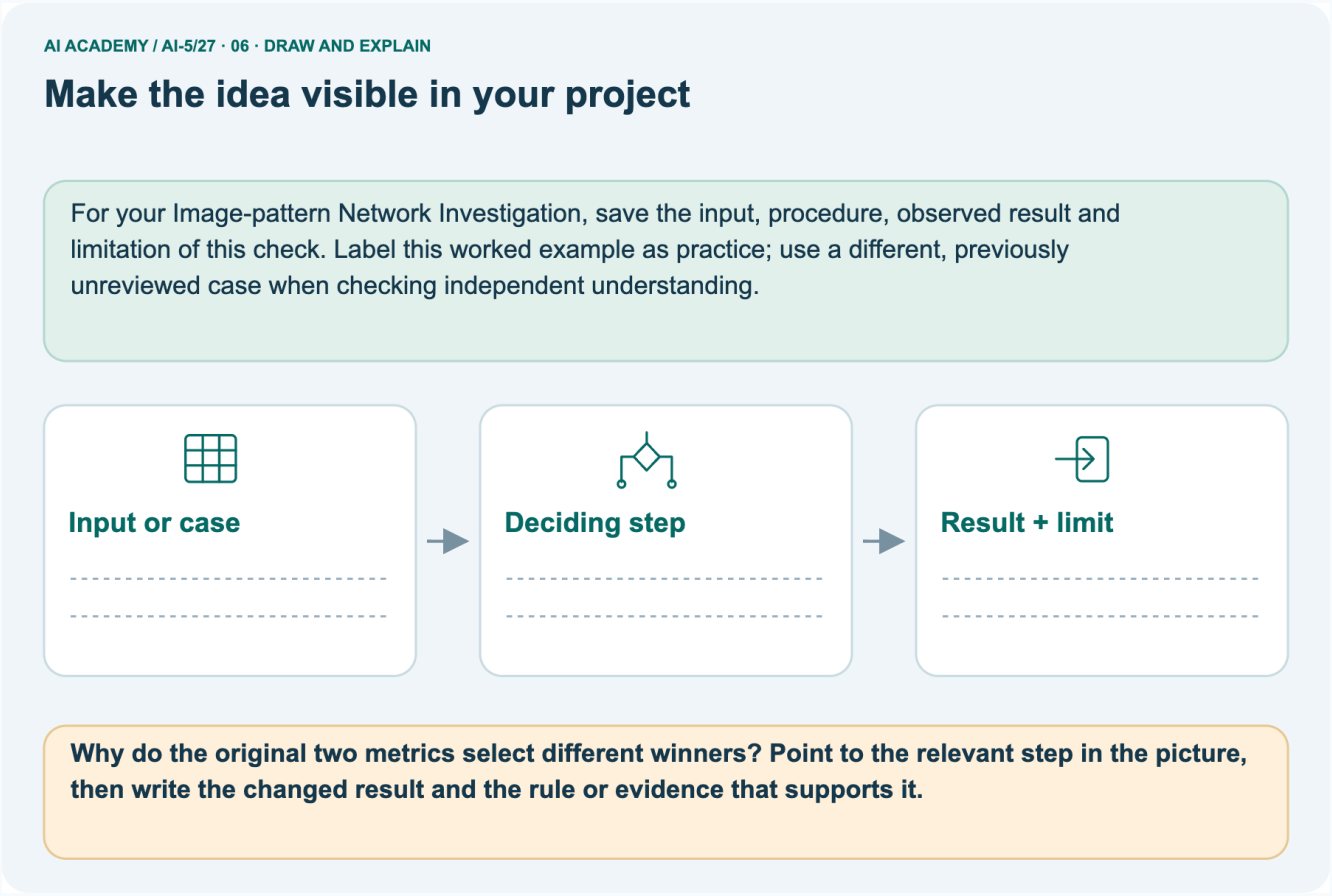
Raw Euclidean distance includes differences in both direction and magnitude. Cosine compares direction using the **vector length**s, so a long vector can align perfectly while being far away in raw coordinates.

Declare which question your similarity measure answers. A zero vector has no direction and cannot be divided by its norm for cosine or unit normalisation.

Limit: Neither metric is universally right; choose it from the task and handle zero vectors explicitly.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why do the original two metrics select different winners? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.